Investigations of irrationality constructions for $\beta(4)$

Technical supplement: obstructions, reductions, and unresolved arithmetic

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Abstract

An accurate rational approximation to a constant need not help prove its irrationality: multiplying the error by the denominator may make it large. This supplement examines that difficulty for $\beta(4) = 1 - 3^{-4} + 5^{-4} - \cdots$ through several natural constructions. It proves failure results for specified polynomial integral sources, asymptotic corrections to partial sums, differentiated continued fractions, and algebraic approximations built from roots of unity. Further arguments identify limitations imposed by positivity constraints and by analytic continuation. The conclusions concern the stated families and operations. Two results explain the limits of a broader negative conclusion: unrestricted positive polynomial sources can represent every positive rational linear form in $\beta(4)$, and a parameter-dependent moment construction gives positive errors with a sharp decay rate while its denominator growth remains unresolved. Selected complete proofs are followed by a guide to the wider investigation, distinguishing proved obstructions, insufficient estimates and finite computations. The companion paper develops two determinant obstructions in detail. Neither document settles the rationality or irrationality of $\beta(4)$.

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1 Introduction and guide to the results

Put $\theta = \beta(4) = \sum_{k\geq 0}(-1)^k/(2k+1)^4$. Suppose a construction gives a rational approximation a_n/q_n in lowest terms, with error $e_n = \theta - a_n/q_n$. The associated integer linear form is $q_n\theta - a_n = q_n e_n$. To prove irrationality by this route, one needs these forms to be nonzero and tend to zero. Indeed, if $\theta = a/b$ were rational, every such nonzero form would have absolute value at least $1/b$. Thus the central comparison is between the decay of e_n and the growth of its reduced denominator.

This supplement asks how that comparison works in several explicit constructions. Some start from integrals, some improve the alternating series by tail corrections or continued fractions, and some use algebraic approximations or differential equations. The companion paper [1] treats two determinant families in detail. Here we give selected further proofs and retain the other attempts with their precise mathematical status.

The results are organized around three questions: what makes the constructed value nonzero, how small is it as a real number, and what arithmetic normalization is unavoidable? To prove failure by comparing sizes, one must show that the last cost defeats the real decay. A large convenient denominator, or a finite run of unfavorable examples, is insufficient for such a conclusion.

The motivating determinant work of OpenAI [11, Section 1.2] separates nonvanishing, real size and arithmetic normalization. Zudilin [13] gives the broader positive-moment determinant framework. The distinction between real decay and primitive arithmetic remained decisive in this investigation. Carreira's work [4] provides directly related $\beta(4)$ constructions and tests; the general positive-weight and rationally modified moment frameworks also appear in Prévost [12] and Krattenthaler [7]. These are sources of context and specific attributed constructions, not inputs asserting a global irrationality theorem.

For a nonzero $Q \in \mathbb{Q}[X]$, write $\mathrm{Prim}(Q) = \kappa Q$, where $\kappa > 0$ and the resulting integer coefficients have gcd one. Then

$$v_p(\kappa) = -\min_j v_p([X^j]Q). \tag{1.1}$$

Here v_p counts powers of the prime p in the numerator minus those in the denominator of a nonzero rational number; $v_p(0) = +\infty$. For example, $v_p(p^{-3}) = -3$. The identity follows because the minimum valuation of a primitive integer coefficient list is zero at every prime. Multiplying Q by a nonzero rational scalar preserves its primitive representative up to sign. A lower bound on a convenient clearing integer is therefore not automatically a lower bound on κ. For example, 5 clears $(6X + 12)/5$, whose primitive multiplier is $5/6$.

If $\theta = a/b$ and $Z \in \mathbb{Z}[X]$ has degree at most d, then $b^d Z(\theta) \in \mathbb{Z}$. Thus nonzero values satisfying $|Z_n(\theta)|^{1/d_n} \to 0$ would prove irrationality. Merely having $Z_n(\theta) \to 0$ with unbounded degrees is insufficient: X^n at $1/2$ is a counterexample. For linear forms this reduces to the usual nonzero integer-form criterion.

A guide to the proved results. The table summarizes the selected statements proved in Sections 2–9. Each section introduces its construction and states its hypotheses. In particular, the continued-fraction conclusion fixes the endpoint and uses even depth, and the Mellin inequality concerns constrained nonnegative sources.

Construction	What is proved here	Section
Positive polynomial integrals	Every positive form $A\theta + B$, $A, B \in \mathbb{Q}$, is representable; an obstruction requires restrictions on the sources.	2
Repeated quadratics	Unavoidable prime denominators defeat the best real decay within the stated family.	3
Euler–Boole corrections	Reduced integer forms grow for every escaping endpoint and truncation schedule.	4
Differentiated continued fractions	A fixed-endpoint, even-depth family has growing reduced forms.	5
Roots of unity	Large conjugates defeat the accuracy of one algebraic approximation.	6
Completed Mellin source	Positivity and two moment constraints force a lower bound after normalization.	7
Linear regularization	Removing a prescribed branch also removes the constant to be tested.	8
Moment differentiation	The error is positive with a sharp decay rate; the denominator question remains open.	9

For a first reading, Sections 3–4 give direct examples of the real-size versus denominator comparison. Section 9 shows what remains when the analytic part succeeds but the arithmetic part is unresolved. The other sections can then be read by method; the integral notation in Section 2 is also used in Section 3.

Four conclusions must be kept separate. A *primitive obstruction* bounds actual primitive values. An *estimate limitation* says only that a specified bound or normalization is inadequate. A *structural obstruction* disproves a particular proposed implication or transformation. *Finite evidence* concerns only the enumerated instances. The guide to the wider investigation in Section 10 preserves these distinctions, including where a later correction weakened an earlier interpretation. Stable labels such as S03 or P07 point to the source register in Appendix A and the 57-entry repository inventory. They identify records, not additional theorems proved here. The organization is by construction, rather than by the chronology of eighteen research rounds.

Standard background includes the prime number theorem $\vartheta(x) = \sum_{p \le x} \log p = x + o(x)$, elementary complex integration, Fourier convergence and the stated elementary polynomial identities. Throughout, $\chi_4(m)$ is zero for even m and $(-1)^{(m-1)/2}$ for odd m. The completed Mellin application has an additional, explicitly identified polylogarithmic input in Section 7. The source register distinguishes proofs rewritten here from results retained as archived claims. No new independent audit of this assembled supplement is asserted.

2 Positive polynomial sources: a scope boundary

Multiplying an integral representation of θ by a nonnegative polynomial is a natural way to preserve the sign while reducing the integral's size. Before restricting to a particular sequence of polynomials, we ask how large a class of rational linear forms this procedure can represent. The answer is every positive one. Thus positivity alone cannot explain failure of this approach; the restrictions on a chosen polynomial family matter.

Define

$$\mathrm{d}\nu(t) = \frac{(-\log t)^3}{96\sqrt{t}}\,\mathrm{d}t, \qquad I(P) = \int_0^1 \frac{P(t)}{1+t}\,\mathrm{d}\nu(t). \tag{2.1}$$

Here $\nu_j = \int t^j \, \mathrm{d}\nu = (2j+1)^{-4}$ and $I(1) = \theta$. For a rational polynomial define its *formal pair* by

$$A(P) = P(-1), \qquad B(P) = \int_0^1 \frac{P(t) - P(-1)}{1+t} \, \mathrm{d}\nu(t). \tag{2.2}$$

The quotient is a polynomial, so $B(P) \in \mathbb{Q}$ and $I(P) = A(P)\theta + B(P)$. This defines the coefficients without assuming that 1 and θ are linearly independent.

Theorem 2.1 (The unrestricted positive cone; S01)**.** *The formal-pair image of $P \in \mathbb{Q}[t]$ nonnegative on $[0,1]$ is*

$$\{(0,0)\} \ \cup \ \{(A,B) \in \mathbb{Q}^2 : A\theta + B > 0\}.$$

The same positive primitive integer pairs are obtained if the source polynomials are required to have integer coefficients.

Proof. Nonzero nonnegative polynomials have strictly positive integrals. Conversely let $L = A\theta + B > 0$. If $A = 0$, use $P = B(1+t)$, since $\nu_0 = 1$. For $A > 0$ use $P_n = A((1-t)/2)^{2n}$; for $A < 0$ use $P_n = (-A)t((1-t)/2)^{2n}$. Both have $A(P_n) = A$, are nonnegative, and satisfy $I(P_n) \le |A|\theta 4^{-n}$. Choose n making this less than L. Then

$$P = P_n + (B - B(P_n))(1+t)$$

is nonnegative and has exactly the desired pair; the added coefficient is rational and equals $L - I(P_n) > 0$. Clearing the rational coefficients of P multiplies its pair by a positive integer. Primitive normalization returns the original primitive pair. □

This construction encodes an already specified approximation; it does not produce a new one. In particular, shrinking positive primitive integer forms in this unrestricted cone exist if and only if θ is irrational. Necessity follows from the rational lower bound $1/b$; sufficiency follows from rational approximation to an irrational real number and Theorem 2.1. A negative theorem for all such sources would therefore be a rationality theorem, not a consequence of failures of a few prescribed weights.

Proposition 2.2 (Cost in degree; S01)**.** *Let $\rho = 3 + 2\sqrt{2}$. For $A \neq 0$ and $L = A\theta + B > 0$, the least degree $d(A,B)$ of a nonnegative rational witness satisfies*

$$d(A,B) = \frac{\log(|A|/L)}{\log \rho} + O\big(\log\log(|A|/L)\big) \qquad (|A|/L \longrightarrow \infty).$$

Proof. Let $q_n(t) = T_n(2t-1)/T_n(-3)$, where T_n is the Chebyshev polynomial. Then $q_n(-1) = 1$ and $\|q_n\|_{[0,1]} \le 2\rho^{-n}$. Replace P_n in the preceding proof by Aq_n^2 or $(-A)tq_n^2$. Its integral is at most $4|A|\rho^{-2n}$ and its degree at most $2n+1$. This gives the upper bound, with an absolute additive constant.

For a degree-D nonnegative P set $M = \max_{[0,1]} P$. Its Chebyshev expansion has constant coefficient at most M in absolute value and other coefficients at most $2M$, by the cosine coefficient integrals. The identity $T_k(\cos u) = \cos(ku)$ gives $\|T_k'\|_{[-1,1]} \le k^2$, hence $\|P'\|_{[0,1]} \le 4MD^3$. On an interval of length $\ell = 1/(8D^3)$ adjoining a maximizer, $P \ge M/2$. On its central half, $1 - t \ge \ell/4$, and $\mathrm{d}\nu/(1+t) \ge (1-t)^3 \, \mathrm{d}t/192$. Thus

$$I(P) \ge \frac{M}{3 \cdot 2^{26} D^{12}}.$$

Interpolation at the $D+1$ Chebyshev extrema gives $|P(-1)| \le MT_D(3) \le M\rho^D$: the exterior Lagrange coefficients alternate in sign, and their absolute sum is $T_D(3)$. Therefore

$$I(P) \ge \frac{|P(-1)|}{3 \cdot 2^{26} D^{12} \rho^D}. \tag{2.3}$$

Combining this with the upper construction proves the assertion. □

The archived bounded-degree extremizers and a degree-16 witness for $5719097 - 5783031\theta > 0$ are finite evidence [record S02]. Their role is to check scope: unfavorable boundary examples do not exclude favorable interior forms. The complete bounded-degree boundary formulas are retained in the source record; they are not needed for either proof above.

3 Rational Euler sources and repeated quadratics

One way to build a small linear form is to cancel many initial terms of a generating series, leaving a rapidly decaying tail. A rational correction is attractive because integration by parts can turn its contribution into a rational boundary value. The Euler operator below multiplies t^k by k; applying it four times matches the fourth powers in the beta series. We first reduce such a construction to a polynomial integral, then test one natural repeated-factor family. The obstruction comes from fourth powers of primes that survive in the reduced form.

Let $E = t\,\mathrm{d}/\mathrm{d}t$, $g(t) = t/(1+t^2)$, and

$$J(R) = \frac{1}{6}\int_0^1 (-\log t)^3 R(t)\frac{\mathrm{d}t}{t}.$$

If H is rational, regular on $[0,1]$, and $H(t) = O(t)$ at zero, four integrations by parts, after $t = e^{-z}$, give

$$J(E^4 H) = H(1), \qquad J(Ag + E^4 H) = A\theta + H(1). \tag{3.1}$$

This isolates the target exactly, but imposes no favorable arithmetic bound on the coefficient A.

Put $F(y) = \sum_{k\ge 0}(-1)^k y^k/(2k+1)^4$. Let $Q(y) = \sum_{j=0}^d q_j y^j$ have no zero on $[0,1]$, and let C_m be the Taylor polynomial of QF through degree $m-1$. The choice $H(t) = -AtC_m(t^2)/Q(t^2)$ cancels the first m odd coefficients of $Ag + E^4 H$. Define

$$W_{Q,m}(y) = \sum_{j=m}^{d} q_j + \sum_{j=0}^{\min(m-1,d)} q_j (-y)^{m-j}.$$

The finite geometric remainder for each monomial proves the exact formal-pair identity

$$\theta - \frac{C_m(1)}{Q(1)} = \frac{I(W_{Q,m})}{Q(1)}. \tag{3.2}$$

This reduction includes $m < d$ and introduces no period-independence assumption. For $Q = (v+uy)^n$ and $m = 2n$, $W_{Q,2n} = (-y)^n(u - vy)^n$. The rational Euler source therefore has the same primitive pair, up to sign, as the following scalar source.

Theorem 3.1 (Repeated-quadratic sources; S03)**.** *For even n, let u_n, v_n be positive integers with $\log(u_n + v_n) = o(n)$. Set $P_n(y) = y^n(u_n - v_n y)^n$ and let $L_n > 0$ be the value of its primitive formal pair under (2.2). Then*

$$\liminf_{\substack{n\to\infty \\ n \text{ even}}} \frac{\log L_n}{n} \ge \frac{8}{3} - \log(5 + 4\sqrt{2}) > 0.$$

Proof. Write $A = P_n(-1) = (u+v)^n$ and $I(P_n) = A\theta + N/D$ in lowest terms. The first n coefficients of $(P_n - A)/(1+y)$ are $(-1)^{j+1}A$. For a prime

$$\frac{4n}{3} < p < 2n, \qquad p \nmid (u+v),$$

only the denominator p among $1, 3, \ldots, 4n-1$ is divisible by p. Its index $(p-1)/2 < n$ has a p-unit coefficient. Consequently $v_p(N/D) = -4$, exactly. If $g = \gcd(DA, N)$, then $g \mid A$, so

$$L_n = \frac{D}{g} I(P_n) \geq \prod_{\substack{4n/3<p<2n \\ p\nmid(u+v)}} p^4 \, \frac{I(P_n)}{A}. \tag{3.3}$$

This controls the actual primitive value, including all common content.

Let $r = v/(u+v)$ and $f_r(y) = y(1-r-ry)$. With $\mathrm{d}\mu = \mathrm{d}\nu/(1+y)$, $I(P_n)/A = \int |f_r(y)|^n \, \mathrm{d}\mu(y)$. An elementary minimax calculation gives

$$M_* := \min_{0\leq r\leq 1} \max_{0\leq y\leq 1} |f_r(y)| = \frac{1}{5+4\sqrt{2}}.$$

Indeed for $r \geq 1/3$ the interior positive maximum is $(1-r)^2/(4r)$; for $r \geq 1/2$ the negative endpoint has magnitude $2r-1$. They are equal at $r = (1+2\sqrt{2})/7$. For $r < 1/3$ the maximum is at least $1/3$.

Uniformity for moving r follows from $|f_r'| \leq 2$. In an interval of length $\ell = M_*/(4n)$ adjacent to a maximizer, $|f_r| \geq M(r)(1 - 1/(2n))$. On its central half, $1 - y \geq \ell/4$ and $\mathrm{d}\mu \geq (1-y)^3 \, \mathrm{d}y/192$. Since $(1 - 1/(2n))^n \geq 1/2$,

$$\frac{I(P_n)}{A} \geq \frac{M_*^{n+4}}{3 \cdot 2^{22} n^4}.$$

The prime number theorem in (3.3) now gives

$$\log L_n \geq \left(\frac{8}{3} - \log(5+4\sqrt{2})\right) n - 4 \sum_{\substack{4n/3<p<2n \\ p\mid u+v}} \log p + o(n).$$

The exceptional sum is at most $\log(u+v) = o(n)$. □

It suffices more generally that the displayed exceptional prime sum be $o(n)$. The theorem does not cover odd n, different osculation lengths, general denominator polynomials, or sums of these forms. The source-level statement is weaker: if $H = P/Q$ has integer coefficients and $p \nmid Q(0)$, cancellation of the first m odd coefficients forces $v_p(A) \geq 4 \max_{k<m} v_p(2k+1)$. This follows coefficientwise from $A(-1)^k + (2k+1)^4 [t^{2k+1}]H = 0$. It does not by itself control primitive endpoint content; that was the purpose of the prime argument.

4 Euler–Boole truncations at every endpoint

The alternating partial sums can be improved by adding an asymptotic expansion of their tails. There are two choices: where to stop the original series and how many correction terms to retain. The best real error in this family is exponentially small, on the scale $e^{-\pi N}$. However, the reduced denominators grow at rate at least $e^{8N+o(N)}$, uniformly in the number of corrections. The theorem below compares these quantities and also handles a fixed endpoint with an increasing number of correction terms.

The Euler–Boole method is classical; the point here is the uniform arithmetic conclusion for this particular family. Define integers E_{2j} by $\operatorname{sech} t = \sum_{j\ge 0} E_{2j} t^{2j}/(2j)!$ near zero. Multiplication by $\cosh t$ gives their integral recurrence. For $N \ge 1$ and $m \ge 0$ put

$$\begin{aligned} S_N &= \sum_{k<N} \frac{(-1)^k}{(2k+1)^4}, \\ T_{N,m} &= \frac{1}{12} \sum_{j<m} \frac{E_{2j}(2j+1)(2j+2)(2j+3)}{(2N)^{2j+4}}, \\ A_{N,m} &= S_N + (-1)^N T_{N,m} = a_{N,m}/q_{N,m}, \end{aligned} \tag{4.1}$$

in lowest terms with $q_{N,m} > 0$. The Euler-number convention and classical summation formula are recorded in DLMF [5, Sections 24.2, 24.17]; the needed remainder is derived below.

Theorem 4.1 (Uniform primitive obstruction; S04)**.** *Every $q_{N,m}\theta - a_{N,m}$ is nonzero. Uniformly in $m \ge 1$,*

$$|\theta - A_{N,m}| \ge \frac{\pi^2 e^{-\pi N}}{144N}. \tag{4.2}$$

Moreover

$$\liminf_{N\to\infty} \inf_{m\ge 0} \frac{\log q_{N,m}}{N} \ge 8, \qquad \liminf_{N\to\infty} \inf_{m\ge 0} \frac{\log |q_{N,m}\theta - a_{N,m}|}{N} \ge 8 - \pi > 0.$$

The primitive absolute values tend to infinity whenever $N + m \to \infty$.

Proof. Laplace integration of the alternating tail gives

$$\theta - S_N = \frac{(-1)^N}{12} \int_0^\infty t^3 e^{-2Nt} \operatorname{sech} t \, \mathrm{d}t. \tag{4.3}$$

Let $a = \pi/2$ and $a_k = (2k+1)a$. The partial fractions formula

$$\operatorname{sech} t = 2 \sum_{k\ge 0} \frac{(-1)^k a_k}{a_k^2 + t^2}$$

follows by expanding $\cosh(tx)/\cosh t$ on $[0,1]$ in the mixed cosine basis $\cos(a_k x)$: twice integrating its differential equation gives coefficient $2a_k(-1)^k/(t^2 + a_k^2)$, and Fourier convergence at $x = 0$ gives the formula. Applying the finite geometric identity to each term gives, for $m \ge 1$,

$$\begin{aligned} \operatorname{sech} t - \sum_{j<m} \frac{E_{2j} t^{2j}}{(2j)!} &= (-1)^m t^{2m} C_m(t), \\ C_m(t) &= 2 \sum_{k\ge 0} \frac{(-1)^k a_k^{1-2m}}{t^2 + a_k^2}. \end{aligned} \tag{4.4}$$

This remainder series is absolutely convergent. Its positive terms decrease, and the ratio of the second to the first is at most $3^{1-2m} \le 1/3$. Hence

$$(1 - 3^{1-2m}) \frac{2a^{1-2m}}{t^2 + a^2} \le C_m(t) \le \frac{2a^{1-2m}}{t^2 + a^2}, \qquad C_m(t) > 0. \tag{4.5}$$

Thus $\theta - A_{N,m} = (-1)^{N+m} \epsilon_{N,m}$ with $\epsilon_{N,m} = \int_0^\infty t^{2m+3} e^{-2Nt} C_m(t) \, \mathrm{d}t/12 > 0$. On $t \ge a$, use $(t/a)^{2m} \ge 1$ and $t^3/(t^2 + a^2) \ge a/2$. The lower bound becomes $\epsilon_{N,m} \ge (\pi/18)(a/2) e^{-2Na}/(2N)$, proving (4.2). For $m = 0$, $\operatorname{sech} t \ge e^{-t}$ gives $|\theta - S_N| \ge 1/[2(2N+1)^4]$.

For the arithmetic, let p be odd, $p^2 > 2N$ and $p < 2N$, and let J count the positive odd r with $pr < 2N$. Reduction yields

$$p^4 S_N \equiv \chi_4(p) S_J \pmod{p}, \qquad \chi_4(p) = (-1)^{(p-1)/2}. \tag{4.6}$$

If $p \nmid 6N$, the correction $T_{N,m}$ is p-integral for every m. Whenever S_J is a nonzero residue, therefore, $v_p(q_{N,m}) = 4$ exactly. Fix $K \geq 1$. For all sufficiently large N, primes in $(2N/(2K+1), 2N)$ have $1 \leq J \leq K$, satisfy $p^2 > 2N$, and avoid the finitely many numerator and denominator prime divisors of the positive rationals $S_1, \ldots, S_K$. Uniformly in m,

$$\log q_{N,m} \geq 4\{\vartheta(2N) - \vartheta(2N/(2K+1)) - \log(6N)\} - O_K(\log N).$$

First take the lower limit in N, then let $K \to \infty$; no uniformity in K is required. This gives the rate 8, and combining with the error estimates gives $8 - \pi$.

For bounded N and $m \to \infty$, restrict the positive remainder integral to $[b, b+1]$, where $b > a$. Its lower bound is $c_{N,b}(b/a)^{2m} \to \infty$, uniformly over a finite set of N. Since $q \geq 1$, this also proves divergence on every escaping parameter sequence. □

The raw exponential constant is sharp: for $m = \pi N/2 + O(1)$,

$$|\theta - A_{N,m}| \sim \frac{\pi^3}{48\sqrt{2N}} e^{-\pi N}. \tag{4.7}$$

Indeed (4.5) reduces the integral to its first pole with relative error $O(3^{1-2m})$. The exponent is $N(2a\log(t/a) - 2t)$, with unique maximum at $t = a$, second derivative $-2/a$, and remaining factor $(\pi/12)t^3/(t^2+a^2)$ there equal to $\pi a/24$. Gaussian localization gives (4.7), uniformly for the bounded rounding error in m.

There are finitely many small primitive values, for example those of $X - 1$, $81X - 80$ and $32X - 31$. They do not contradict divergence. Nonvanishing and divergence off finite parameter sets give $c := \inf_{N,m} |q_{N,m}\theta - a_{N,m}| > 0$. By Gauss's lemma a product of d of these primitive linear polynomials is primitive, and its dth-root absolute value is at least c. Products alone therefore cannot supply the degree-normalized smallness criterion. Sums, resultants and general determinant combinations are outside this result.

5 Differentiated continued fractions at fixed endpoints

Continued fractions give rational approximations to a tail integral without truncating its asymptotic series. Here three derivatives with respect to the Laplace parameter introduce the factor t^3 needed for the fourth-power beta tail. The resulting approximation has an explicit error asymptotic, but differentiating a quotient also raises powers of its denominator. For a fixed tail endpoint and increasing even depth, the error decays only polynomially while the reduced denominator grows exponentially. The proof checks this cost after all cancellation.

For $\Re a > 0$ let $S(a) = \int_0^\infty e^{-at} \operatorname{sech} t \, \mathrm{d}t$. The numerator and denominator polynomials of the approximants, called continuants, are defined by

$$q_0 = 1,\ q_1 = a, \quad r_0 = 0,\ r_1 = 1, \qquad (n+1)y_{n+1} = ay_n + ny_{n-1}.$$

The integer polynomials $Q_n = n!q_n$, $R_n = n!r_n$ satisfy $Y_{n+1} = aY_n + n^2 Y_{n-1}$. Their quotient is the finite continued fraction with successive numerators $1^2, 2^2, \ldots$. Its convergence will follow from an explicit error, rather than an imported convergence theorem. Put

$$\alpha_{n,N} = S_N + \frac{(-1)^{N+1}}{12} \frac{\mathrm{d}^3}{\mathrm{d}a^3} \frac{r_n(a)}{q_n(a)}\bigg|_{a=2N} = \frac{b_{n,N}}{d_{n,N}}, \tag{5.1}$$

in lowest terms, $d_{n,N} > 0$. Here S_N is the finite sum in (4.1). The entire quotient is differentiated.

Theorem 5.1 (Fixed-endpoint obstruction; S05)**.** *For each fixed integer $N \geq 1$,*

$$\theta - \alpha_{n,N} \sim (-1)^{N+n} \frac{2^{-2N}\Gamma(N+\frac{1}{2})^2}{12} n^{-2N} (\log n)^3. \tag{5.2}$$

Along even depth $n = 2m$,

$$\liminf_{m\to\infty} \frac{\log |d_{2m,N}\theta - b_{2m,N}|}{2m} \geq 4.$$

No uniformity for growing N, or arithmetic conclusion at odd depth, is part of this theorem.

5.1 The exact error and its derivative

The generating functions of the normalized continuants are

$$\mathcal{Q}(a,z) = (1+z)^{(a-1)/2}(1-z)^{-(a+1)/2},$$
$$\mathcal{R}(a,z) = \mathcal{Q}(a,z) \int_0^z \frac{\mathrm{d}t}{(1-t^2)\mathcal{Q}(a,t)}.$$

They follow by multiplying the recurrence by z^n and integrating the resulting first-order equation. Substitution $z = \tanh u$ and translation of the tail integral give

$$\mathcal{Q}(a,z)S(a) - \mathcal{R}(a,z) = \int_0^\infty \frac{e^{-av}}{\cosh v + z \sinh v}\,\mathrm{d}v.$$

Expanding on compact subdisks $|z| < 1$ proves

$$q_n(a)S(a) - r_n(a) = (-1)^n I_n(a), \qquad I_n(a) = \int_0^\infty e^{-at} \operatorname{sech} t \tanh^n t\,\mathrm{d}t. \tag{5.3}$$

Since $\theta = S_N + (-1)^{N+1} S'''(2N)/12$,

$$\theta - \alpha_{n,N} = \frac{(-1)^{N+n+1}}{12} \left.\frac{\mathrm{d}^3}{\mathrm{d}a^3} \frac{I_n(a)}{q_n(a)}\right|_{a=2N}.$$

Uniformly on a small fixed complex disk about $2N$ with $\Re a > 1$,

$$q_n(a) = \frac{2^{(a-1)/2} n^{(a-1)/2}}{\Gamma((a+1)/2)}(1 + O_N(n^{-1})),$$
$$I_n(a) = 2^{-(a+1)/2}\Gamma((a+1)/2) n^{-(a+1)/2}(1 + O_N(n^{-1})).$$

For the first formula, expand the displayed generating function at $z = 1$ and use the coefficient formula for $(1-z)^{-b}$; the second singularity $z = -1$ has relative size $O(n^{-\Re a})$. For the second, put $y = \tanh t$ and expand $(1+y)^{-(a+1)/2}$ at $y = 1$ in

$$I_n(a) = \int_0^1 y^n (1-y)^{(a-1)/2}(1+y)^{-(a+1)/2}\,\mathrm{d}y.$$

The beta integral and gamma-ratio estimate give the stated constants. The remainders are holomorphic and uniform on a slightly larger disk; the leading factor of q_n is nonzero there for large n. Cauchy's derivative formula therefore justifies three derivatives of $2^{-a}\Gamma((a+1)/2)^2 n^{-a}(1+O_N(n^{-1}))$. Its leading third derivative is the negative of its leading term times $(\log n)^3$, proving (5.2). The standard gamma-ratio expansion is also recorded in DLMF [5, Section 5.11(iii)]. Pointwise real asymptotics without this complex uniformity would not justify the differentiation.

5.2 An exact reduced-denominator law

Write $X^{\underline{j}} = X(X-1)\cdots(X-j+1)$ and $h_N = (2N-1)!!$. Define integer polynomials

$$\begin{aligned}
A_N(X) &= \sum_{j=0}^{N} \binom{2N}{2j} 2^j X^{\underline{j}} \prod_{\ell=0}^{N-j-1} (2X+2\ell+1),\\
B_N(X) &= \sum_{j=0}^{N-1} \binom{2N}{2j+1} 2^j X^{\underline{j}} \prod_{\ell=0}^{N-j-1} (2X+2\ell+1),\\
C_N(m) &= B_N(m-\tfrac{1}{2})\\
&= \sum_{j=0}^{N-1} \binom{2N}{2j+1} \prod_{s=0}^{j-1}(2m-1-2s) \prod_{\ell=0}^{N-j-1} (2m+2\ell).
\end{aligned}$$

The last expression is integral for integer m, positive for $m \geq N$, and of fixed degree N. With $c_m = \binom{2m}{m}/4^m$,

$$q_{2m}(2N) = c_m A_N(m)/h_N, \qquad q_{2r+1}(2N) = c_r B_N(r)/h_N. \tag{5.4}$$

To check these, write $\mathcal{Q}(2N,z) = (1+z)^{2N}(1-z^2)^{-N-1/2}$ and separate even and odd coefficients. The coefficient ratio used is

$$\frac{(N+\frac{1}{2})_{m-j}/(m-j)!}{c_m} = \frac{m^{\underline{j}}(m+\frac{1}{2})_{N-j}}{(\frac{1}{2})_N}.$$

Lemma 5.2. *If $m < p < 2m$, $2N < p$, p is prime and $p \nmid C_N(m)$, then*

$$v_p(d_{2m,N}) = 8 + 4v_p(A_N(m)).$$

Proof. First $Q_p(a) \equiv a^p - a \pmod p$. One direct proof uses $D = (1-z^2)\mathrm{d}/\mathrm{d}z + (p-1)z$ on polynomials of degree at most $p-1$ over $\mathbb{F}_p$. Adjacent off-diagonal products in its matrix are $j(p-j)$, so its characteristic determinants obey the continuant recurrence modulo p. Its eigenvectors $(1+z)^j(1-z)^{p-1-j}$ have distinct eigenvalues $2j-(p-1)$, which run through $\mathbb{F}_p$. The recurrence then gives

$$Q_{p+j}(a) \equiv (a^p - a)Q_j(a) \pmod p \quad (0 \leq j < p).$$

The continuant Wronskian is $Q_k R_{k+1} - Q_{k+1}R_k = (-1)^k(k!)^2$. At $k = p-1$, $a = 2N$, it shows that $R_p(2N)$ is a unit. Recurrence after p and differentiation of the last congruence give

$$R_{p+j}(2N) \equiv R_p(2N)Q_j(2N), \qquad Q'_{p+j}(2N) \equiv -Q_j(2N) \pmod p.$$

Here $j = 2m-p$ is odd, and (5.4) shows that $Q_j(2N)$ is a unit precisely when $B_N(m-(p+1)/2) \equiv C_N(m)$ is. Thus $R_{2m}(2N)$ and $Q'_{2m}(2N)$ are units. Meanwhile (5.4) gives

$$e := v_p(Q_{2m}(2N)) = 2 + v_p(A_N(m)),$$

because both $(2m)!$ and c_m have valuation one.

Expand the integer polynomials as $Q(2N+x) = \sum u_i x^i$ and $R(2N+x) = \sum v_i x^i$. The coefficient of x^3 in R/Q is

$$\frac{v_3u_0^3 - v_2u_1u_0^2 + v_1(u_1^2u_0 - u_2u_0^2) + v_0(-u_1^3 + 2u_1u_2u_0 - u_3u_0^2)}{u_0^4}.$$

Its numerator has the unique unit term $-v_0u_1^3$, so its valuation is $-4e$. Formula (5.1) multiplies this Taylor coefficient by $\pm 1/2$; S_N is p-integral. It follows that the reduced denominator has valuation $4e$, as asserted. □

For fixed N, primes in $(m, 2m)$ excluded by $C_N(m)$ have total logarithm $O_N(\log m)$. The lemma and the prime number theorem imply $\log d_{2m,N} \geq 8m + o(m)$. Equation (5.2) supplies an eventually nonzero error with logarithm $O_N(\log m)$, completing Theorem 5.1. The exception condition is material: four archived cases excluded by $C_N(m)$ have denominator exponent four, not eight. None of this extends automatically to N growing with m.

A shift-and-binomial repair has an exact reduction: combining the error integrals at $a = 2(L + j)$ uses

$$e^{-2Lt} \tanh^n t (1 + e^{-2t})^n = e^{-2Lt}(1 - e^{-2t})^n.$$

After $x = e^{-t}$ this is the scalar Mellin weight $x^{2L}(1 - x^2)^n$. The proposed repair is therefore an old scalar family in a different presentation, not an additional approximation.

6 A sine-normalized cyclotomic family

Algebraic approximations offer another route: choose an algebraic number α close to θ and evaluate its integer minimal polynomial at θ. This evaluation contains a factor $\theta - \alpha$, but also a factor for every other conjugate of α. A good approximation at one real embedding can therefore be defeated by large conjugates. The roots-of-unity family below exhibits this explicitly, including the removal of common factors from the polynomial coefficients.

The topological-signature motivation came from the cubic cotangent formula recorded by Marché and Masbaum [10, Corollary 3.2], who credit its simplification to Murakami. That signature formula is not an even-power integrality theorem and is not used below. The following algebraic and analytic claims are derived directly for one specified normalization.

For an odd prime p, set $h = \pi/(2p)$, $w = e^{2\pi i/(4p)}$, and

$$\alpha_p = \sin^4 h \sum_{\substack{1 \leq n < p \\ n \bmod 4 = 1}} \chi_4(n) \frac{\cos^3(nh)}{\sin^4(nh)}. \tag{6.1}$$

Let $F_p = \mathbb{Q}(w + w^{-1})$, of degree $d = p - 1$, and $M_p(X) = \mathrm{Norm}_{F_p/\mathbb{Q}}(X - \alpha_p)$. Here the norm polynomial is the product of $X - \sigma(\alpha_p)$ over all embeddings $\sigma : F_p \hookrightarrow \mathbb{R}$.

Theorem 6.1 (One cyclotomic norm obstruction; S06)**.** *The number $8\alpha_p$ is integral, α_p has degree $p - 1$, and its primitive minimal polynomial P_p satisfies*

$$\log |P_p(\theta)| = 4(p - 1) \log p + O(p).$$

In particular these primitive evaluations grow along the odd primes.

Proof. The identity

$$8\alpha_p = \sum_{\substack{n < p \\ n \bmod 4 = 1}} \chi_4(n) \left(\frac{w - w^{-1}}{w^n - w^{-n}} \right)^4 (w^n + w^{-n})^3$$

proves integrality. Indeed $z = w^2$ is a primitive $2p$th root and $1 - z^n$ is a cyclotomic unit for odd $0 < n < p$, since $\Phi_{2p}(1) = 1$. Polynomial division of $\Phi_{2p}(x) - \Phi_{2p}(1)$ by $x - 1$ proves the unit assertion. The displayed expression is real and belongs to F_p.

The principal approximation has the signed expansion

$$\alpha_p = \theta - \frac{\pi^2(4\theta + 5\beta(2))}{24p^2} + O(p^{-4}). \tag{6.2}$$

To prove it, write $\cos^3 x/\sin^4 x = x^{-4} - (5/6)x^{-2} + R(x)$, where R is smooth on $[0, \pi/2]$. Alternating samples of R have uniformly bounded sum by its total variation, and the alternating tails of n^{-4} and n^{-2} are bounded by p^{-4} and p^{-2}. Multiply by $\sin^4 h$ and use $(\sin h/h)^4 = 1 - (2/3)h^2 + O(h^4)$. The coefficient of p^{-2} is strictly positive.

Represent the real embeddings by odd $1 \le k < 2p$, $k \ne p$. The conjugate $\alpha_{p,k}$ is given by (6.1) with each angle multiplied by k. Let

$$f(x) = \frac{\cos^3 x}{\sin^4 x}, \qquad C = \sum_{n \ge 1,\ n\ modd} n^{-4} = \frac{\pi^4}{96}, \qquad \delta = 2 - C > 0.$$

The function $x^4 f(x)$ strictly decreases on $(0, \pi/2)$. The sign of its logarithmic derivative reduces to positivity of $H(x) = x(3 + \cos^2 x) - 4 \sin x \cos x$; indeed $H(0) = 0$ and $H'(x) = 7 \sin^2 x - 2x \sin x \cos x \ge 5 \sin^2 x$. Thus $f(mh) \le f(h)/m^4$ for odd $1 \le m < p$.

Multiplication by k permutes the pairs of odd nonzero residues $\{n, 2p-n\}$ modulo $2p$. The inner conjugate sum is therefore $\sum_m \eta_m f(mh)$, where each $\eta_m = \pm 1$ and each odd $0 < m < p$ occurs once. Its first term exceeds the sum of the absolute values of the rest. Uniformly in k,

$$\delta \cos^3 h \left(\frac{\sin(kh)}{\sin h}\right)^4 \le |\alpha_{p,k}| \le C \cos^3 h \left(\frac{\sin(kh)}{\sin h}\right)^4. \tag{6.3}$$

For $p \ge 5$ and $k \notin \{1, 2p-1\}$, the sine ratio is at least $\sin(3h)/\sin h \ge 2$, so these conjugates have absolute value greater than 10 (use $\delta > 0.98$, $\cos^3 h \ge 3\sqrt{3}/8$). The principal conjugate is in $(0, C)$ and the $2p-1$ conjugate is its negative. Hence it is the unique conjugate in $(0, 2)$ and its stabilizer is trivial. The degree is d. For $p = 3$, directly $\alpha_3 = 3\sqrt{3}/8$ has degree two.

The odd sine product is

$$\prod_{\substack{1 \le k < 2p \\ k\ modd,\ k \ne p}} \sin(kh) = 2^{1-p} = 2^{-d}.$$

It follows by evaluating $x^p + 1$ at $x = 1$ in its root factorization and removing the sine-one factor. Therefore (6.3) implies

$$\left(\frac{\delta \cos^3 h}{16 \sin^4 h}\right)^d \le \prod_k |\alpha_{p,k}| \le \left(\frac{C \cos^3 h}{16 \sin^4 h}\right)^d.$$

For nonprincipal conjugates, replacing $|\alpha_{p,k}|$ by $|\theta - \alpha_{p,k}|$ changes the product logarithm by $O(d)$, since $0 < \theta < 1$ and their magnitudes exceed 10. For the two remaining conjugates, (6.2) gives a total change $-2 \log p + O(1)$. Since $\sin h \sim \pi/(2p)$, $\log |M_p(\theta)| = 4d \log p + O(p)$.

Finally $J_p(X) = \mathrm{Norm}(8X - 8\alpha_p) = 8^d M_p(X)$ is integral, and its content g_p divides its leading coefficient 8^d. Thus $P_p = J_p/g_p$ is the actual primitive minimal polynomial and $|M_p(\theta)| \le |P_p(\theta)| \le 8^d |M_p(\theta)|$. This proves the theorem after all content is removed. □

The norm polynomial is evaluated at the real number θ; no assertion that $\theta - \alpha_p \in F_p$ is made without first assuming θ rational. Under $\theta = a/b$, the integer test is $b^d P_p(a/b) \in \mathbb{Z} \setminus \{0\}$. Its degree cost cannot be omitted. Neither other normalizations nor other nonlinear norms are excluded.

7 A constrained completed Mellin source

Functional identities can express θ through polylogarithms at arguments strictly inside the unit disk, where their series converge geometrically. This suggests a source of accurate approximations. Turning such identities into polynomially weighted integrals, however, introduces lower-order logarithmic

terms. Cancelling those terms imposes two moment constraints. For the resulting kernel, these constraints force a positive lower bound whenever the multiplier polynomial is nonnegative; the hoped-for small positive forms cannot arise in that class.

This section separates an analytic inequality for an explicit rational kernel from its polylogarithmic identification with θ. The latter uses a published identity; it is not a newly discovered functional equation.

7.1 The external identity and the rational kernel

For $|z| < 1$, the polylogarithm is $\mathrm{Li}_s(z) = \sum_{k\ge1} z^k/k^s$. Its relation to the integrals here is

$$\int_0^1 x^{j-1}\frac{(-\log x)^{s-1}}{(s-1)!}\,\mathrm{d}x = j^{-s} \qquad (j, s \ge 1),$$

obtained by $x = e^{-t}$ and integration by parts. A *completion* adds lower-order polylogarithms with logarithmic coefficients to obtain the cancellation required by the identity. In this case set

$$D_4(z) = \Im\left(\mathrm{Li}_4(z) - \log|z|\,\mathrm{Li}_3(z) + \frac{\log^2|z|}{3}\,\mathrm{Li}_2(z)\right).$$

The weight-two, three and four ladder identities of Broadhurst [2, equations (39)–(46), (54)–(55)] give

$$\theta = \frac{225}{136}D_4(i/2) - \frac{25}{34}D_4((1+i)/4) + \frac{9}{68}D_4((1-i)/8). \tag{7.1}$$

For clarity about completion, write P_s, Q_s, R_s for the raw ladders

$$2^{s-1}\Im\,\mathrm{Li}_s((1+i)/2), \quad (2/3)^{s-1}\Im\,\mathrm{Li}_s((1+i)/4)+\Im\,\mathrm{Li}_s(i/2), \quad (2/5)^{s-1}\Im\,\mathrm{Li}_s((1-i)/8)+2\Im\,\mathrm{Li}_s(i/2).$$

With $\ell = \log 2$ and $\widehat{X} = X_4 + \ell X_3 + (\ell^2/3)X_2$, those source identities give $\widehat{Q} - (2/3)\widehat{P} = -(80/27)\theta$ and $\widehat{R} - (4/5)\widehat{P} = -(384/125)\theta$. Their linear combination eliminating $\widehat{P}$ yields (7.1). The $\beta(3)\ell$ and $\beta(1)\ell^3$ terms cancel only after this completion. Broadhurst derives the source identities by specializing Kummer's functional equations. That external derivation is not reproduced or independently certified here. The inequality below is a direct theorem about U and W and does not depend on accepting (7.1); its interpretation as a θ-form uses this explicitly declared input.

Let $r = -\log x$, $L = \log 2$, and

$$\begin{aligned} W(r) &= r^3/6 + Lr^2/4 + L^2 r/12, \\ U(x) &= \frac{900x^2}{17(x^4+4)} - \frac{2025x^3}{17(x^6-4x^3+8)} - \frac{5625x^5}{17(x^{10}-8x^5+32)}. \end{aligned} \tag{7.2}$$

For $m = 2, 3, 5$, respectively, put

$$(z_m) = (i/2, (1+i)/4, (1-i)/8), \qquad (c_m) = (225/136, -25/34, 9/68).$$

Then $U = \sum c_m m^4 \Im[z_m x^m/(1 - z_m x^m)]$. Geometric expansion and the logarithmic moments show that

$$\Theta := \int_0^1 U(x)W(-\log x)\frac{\mathrm{d}x}{x} = \sum_m c_m D_4(z_m) = \theta, \tag{7.3}$$

where only the final equality uses (7.1).

Write $p_2 = x^4 + 4$, $p_3 = x^6 - 4x^3 + 8$, $p_5 = x^{10} - 8x^5 + 32$ and

$$D = \frac{p_2 p_3 p_5}{(x^2 + 2x + 2)^2}, \qquad N = DU.$$

Exact division gives a monic degree-16 denominator D, positive on $[0, 1]$. For $P = A + DV$, with $A \in \mathbb{Q}$, $V \in \mathbb{Q}[x]$, write $NV = \sum_{j \geq 1} a_j x^j$ and impose

$$\sum_j \frac{a_j}{j^3} = 0, \qquad \sum_j \frac{a_j}{j^2} = 0. \tag{7.4}$$

Termwise integration proves

$$I_U(P) := \int_0^1 UPW \frac{\mathrm{d}x}{x} = A\Theta + \sum_j \frac{a_j}{j^4}. \tag{7.5}$$

Without both constraints, there are extra $L/2$ and $L^2/12$ times the two sums; congruence modulo D alone does not isolate θ.

7.2 A positive-source lower bound

Theorem 7.1 (Constrained positive sources; S09)**.** *If $P = A + DV$ is nonzero, rational and nonnegative on $[0, 1]$, and satisfies (7.4), then $A > 0$ and $I_U(P) > 1309A/2600$. Consequently every positive rational normalization making A a nonzero integer gives a value greater than $1/2$. This includes primitive normalization of (7.5).*

Proof. One has $U = 225x^2 f(x)/(17D(x))$, with

$$\begin{aligned} f(x) = {} & 4x^{12} - 25x^{11} + 68x^{10} - 145x^9 + 220x^8 - 232x^7 + 260x^6 \\ & - 464x^5 + 880x^4 - 1160x^3 + 1088x^2 - 800x + 256. \end{aligned}$$

A rational Sturm sequence has degrees $12, 11, \ldots, 0$ and variation counts $7, 7, 6, 6$ at $0, 3/5, 5/8, 1$. The rational certificate accompanying this supplement recomputes the sequence. Also $f(0) = 256$, $f(1) = -50$, $f(3/5) = 802361314/244140625 > 0$, and $f(5/8) = -31248983321/17179869184 < 0$. Thus U changes sign exactly once, at $3/5 < x_0 < 5/8$, and its sign is the sign of $r - r_0$ with $r_0 = -\log x_0$.

Let $M_j = \int_0^1 U(x) r^j \, \mathrm{d}x/x$, $j = 1, 2$. The constraints give $\int UPr^j \, \mathrm{d}x/x = AM_j$. Hence

$$\int_0^1 UP\, r(r - r_0) \frac{\mathrm{d}x}{x} = A(M_2 - r_0 M_1) > 0.$$

The same integral with $P = 1$ proves $M_2 - r_0 M_1 > 0$, so $A > 0$. Now

$$W(r) - (L/4 + r_0/6)r^2 - (L^2/12)r = r^2(r - r_0)/6.$$

Multiplication by UP and integration gives the strict bound $I_U(P) > A[(L/4 + r_0/6)M_2 + (L^2/12)M_1]$.

The exact rational tail estimates described below prove $M_1 > 9/10$ and $M_2 > 193/100$. The elementary inequality $\log a > 2(a - 1)/(a + 1)$ for $a > 1$ gives $L > 2/3$ and $r_0 > \log(8/5) > 6/13$. Therefore the bracket exceeds

$$(1/6 + 1/13)(193/100) + (1/27)(9/10) = 1309/2600 > 1/2.$$

If κA is a nonzero integer and $\kappa > 0$, it is at least one. This proves the normalization conclusion. □

Here the finite certificate has a precise mathematical role. The moment formulas are

$$M_1 = \sum_m c_m m^2 \Im \operatorname{Li}_2(z_m), \qquad M_2 = 2\sum_m c_m m \Im \operatorname{Li}_3(z_m).$$

Use $\rho_m = 1/2, 3/8, 3/16$, respectively. For $s = 2, 3$, the remainder after $K = 32$ Gaussian-rational terms is at most

$$\frac{\rho_m^{K+1}}{(K+1)^s(1-\rho_m)}.$$

All endpoints are rational. Their resulting intervals are contained in

$$(0.9006560116125, 0.9006560116154), \qquad (1.9321066776805, 1.9321066776807),$$

respectively; the exact fractions, not those decimal displays, are used in the checks. Together with the Sturm sequence, this is a finite exact certificate for the theorem's constants.

For signed P, the proof no longer bounds the actual integral. It still gives a limitation on the absolute-integrand estimate:

$$\int_0^1 |UP| W \frac{\mathrm{d}x}{x} \geq \frac{L^2}{12}|A|M_1.$$

After κA is a nonzero integer, the right side exceeds $1/30$. Signed cancellation is not excluded. In fact the archived Rodrigues subspace [record L03] reduces the rational moment denominator to a single lcm, but also contains explicit nonzero sources with zero formal pair. Small source coefficients alone do not settle endpoint nonvanishing.

8 A linear regularization that erases its endpoint constant

The arithmetic-holonomy framework of Calegari, Dimitrov and Tang [3] motivates trying to turn rationality of a constant into a restriction on analytic continuation. Here we examine one preliminary step of such a proposal. Adding a constant makes all Taylor coefficients at one point rational, but leaves a logarithmic branch at another point. Could a linear differential operation remove that branch while retaining the constant we want to test?

The answer for this particular function is no: any such operation also annihilates constants. The proof uses *monodromy*, the change in a function after analytic continuation around a singularity. One circuit followed by a second produces a nonzero additive constant, forcing the conclusion. No general holonomy bound is applied.

Let $E = x\,\mathrm{d}/\mathrm{d}x$ and $F(x) = \sum_{m\geq 1} \chi_4(m) x^m/m^4$. Put $F_j = E^j F$ and

$$C(x) = F_0 - \log x\, F_1 + \frac{\log^2 x}{2} F_2 - \frac{\log^3 x}{6} F_3. \tag{8.1}$$

Since $E^4 F = x/(1+x^2)$, differentiation telescopes:

$$C'(x) = -\frac{\log^3 x}{6(1+x^2)}, \quad C(x) = \int_0^x \frac{-\log^3 t}{6(1+t^2)}\,\mathrm{d}t, \quad C(1) = \theta.$$

For any rational q, $C_q = C + q - \theta$ has a rational Taylor germ at $x = 1$, including its constant term. This is a full rational-jet repair, not just a rational value. It does not remove the logarithmic branch at zero.

Proposition 8.1 (Scoped monodromy obstruction; E04)**.** *Let L be a linear differential operator with rational-function coefficients in x, even over $\mathbb{C}$. If $L(C_q)$ is meromorphic near zero, then $L(1) = 0$. In particular $L(C_q) = L(C)$ is independent of the additive constant $q - \theta$.*

Proof. Let M_0 denote continuation once positively about zero and $\Delta = (M_0 - 1)C_q$. All F_j are holomorphic at zero, and

$$\Delta'(x) = -\frac{(\log x + 2\pi i)^3 - \log^3 x}{6(1 + x^2)}.$$

Meromorphicity implies $L(\Delta) = 0$. Continue to $x = i$ on the branch with $\log i = \pi i/2$. Then

$$\operatorname{Res}_{x=i} \Delta' = -\frac{(5\pi i/2)^3 - (\pi i/2)^3}{12i} = \frac{31\pi^3}{24} \neq 0.$$

A small circuit about i adds $31i\pi^4/12$ to Δ. The coefficients of L are single-valued. Continuing $L(\Delta) = 0$ and subtracting thus gives $(31i\pi^4/12)L(1) = 0$. Paths can avoid the finite set of poles of L. □

The symmetry $C(1/x) = C(x)$ follows by differentiation and equality at $x = 1$. With $x = (1 - z)/(1 + z)$ and $y = z^2$ it yields

$$C(x) = \theta + J(y), \qquad J'(y) = -\frac{2}{3}\frac{yA(y)^3}{1 + y}, \qquad A(y) = \frac{\operatorname{atanh}\sqrt{y}}{\sqrt{y}}, \quad J(0) = 0.$$

Writing $B = yA$ gives the rational system

$$(B^k)' = \frac{kB^k}{2y} + \frac{kB^{k-1}}{2(1 - y)} \ (k = 1, 2, 3), \qquad J' = -\frac{2B^3}{3y^2(1 + y)}.$$

The span of $1, B, B^2, B^3, J$ is unchanged by replacing J with $q + J$. It has dimension five: B has infinitely many monodromy values about 1, so is transcendental over $\mathbb{C}(y)$, while J' has nonzero residue $\pi^3/96$ at $y = -1$, where the first four functions are locally meromorphic. Adding a rational constant changes neither this module nor its continuation domains. These observations explain this proposed shortcut's failure; they do not exclude nonlinear expressions, extra functions or different differential systems.

9 An unresolved model with complete analytic control

Differentiating a parameter-dependent approximation to $\beta(2)$ can produce one for $\beta(4)$: differentiation raises the denominator power in the defining series. The difficulty is to cancel the remaining $\beta(2)$ term and control the sign of the new error. A moment matrix does both here. Its block determinant isolates a rational approximation, and a variational formula shows that the differentiated error is positive. This section therefore reaches the final arithmetic question rather than proving another failure theorem: how quickly do the reduced denominators of these rational approximations grow?

The ordinary-moment first variation [record P07] is retained because its analytic conclusions are stronger than a negative summary of the project might suggest. It has all-index positivity and a sharp scalar error rate. The remaining issue is primitive arithmetic.

For $|u| < 1$ let

$$F(u) = \sum_{k \geq 0} \frac{(-1)^k}{(2k + 1)^2 - u}, \qquad q_j(u) = \frac{1}{(2j + 1)^2 - u}, \qquad w_u(x) = \frac{\sinh(\sqrt{u}(-\log x))}{\sqrt{u}(1 + x^2)}.$$

Use the analytic value of the quotient at $u = 0$. The density is positive for $0 \le u < 1$; it need not be positive for negative u. The moments $m_j(u) = \int_0^1 x^{2j} w_u(x)\,\mathrm{d}x$ obey $m_j + m_{j+1} = q_j$, with $m_0 = F$, $F(0) = \beta(2)$ and $F'(0) = \theta$. These identities follow by expansion and Laplace integration, locally uniformly in u.

For $d = n-1$ set $c_j = q_j$ and $B_{ij} = q_{i+j} + q_{i+j+1}$, $0 \le i, j < d$. Put $a_n = \det B$ and $r_n = c^T B^{-1} c$; for $n = 1$ use $a_1 = 1, r_1 = 0$. In the basis $1, (1+t), (1+t)t, \ldots, t = x^2$, the moment matrix is

$$\begin{pmatrix} F & c^T \\ c & B \end{pmatrix}, \qquad H_n = a_n(F - r_n).$$

The functions $a_n(u)$ and $r_n(u)$ are rational functions with rational coefficients, regular at zero; their values and derivatives there are rational. The following antisymmetric combination of the determinant and its derivative eliminates $\beta(2)$ exactly:

$$W_n = a_n(0)H_n'(0) - a_n'(0)H_n(0) = a_n(0)^2(\theta - r_n'(0)). \tag{9.1}$$

This is a formal identity, with no independence assumption about periods.

Proposition 9.1 (Positivity and the scalar rate; P07)**.** *Let $E_n = \theta - r_n'(0)$ and $\rho = 3 + \sqrt{8}$. Then $E_n > 0$ for every n, and*

$$\lim_{n\to\infty} \frac{\log E_n}{n} = -2\log\rho.$$

Proof. Completing the square in the block matrix gives

$$F(u) - r_n(u) = \min_{\substack{\deg P < n \\ P(-1)=1}} \int_0^1 P(x^2)^2 w_u(x)\,\mathrm{d}x.$$

The unique minimizing polynomial is $P_{n,u}(t) = 1 - (1+t)\sum_{j<d}(B^{-1}c)_j t^j$ and is analytic near zero. Differentiation of the minimum contributes no derivative of P: that term vanishes by stationarity under the constraint. Thus

$$E_n = \int_0^1 P_{n,0}(x^2)^2 \frac{(-\log x)^3}{6(1+x^2)}\,\mathrm{d}x > 0. \tag{9.2}$$

Let $\lambda_{n,s}$ be the analogous constrained minimum for the weight $(-\log x)^{s-1}/[(s-1)!(1+x^2)]$, $s = 2, 4$. Then $E_n \ge \lambda_{n,4}$, while $P_{n,0}$ minimizes the weight-two norm. In the variable $t = x^2$ the densities are

$$\eta_2(t) = \frac{-\log t}{4\sqrt{t}(1+t)}, \qquad \eta_4(t) = \frac{(-\log t)^3}{96\sqrt{t}(1+t)}.$$

The test polynomial $T_{n-1}(2t-1)/T_{n-1}(-3)$ gives $\limsup n^{-1}\log\lambda_{n,s} \le -2\log\rho$.

For the opposite bound restrict to $I_\epsilon = [\epsilon, 1-\epsilon]$. Each density has a positive minimum there. An elementary Legendre evaluation estimate for an interval $I = [a, b]$ of length ℓ is

$$|P(t_*)|^2 \le \frac{n^2}{\ell} R^{2(n-1)} \int_I P(t)^2\,\mathrm{d}t \quad (\deg P < n), \tag{9.3}$$

where $R = 1$ if $t_* \in I$, and otherwise $R = |z_*| + \sqrt{z_*^2 - 1}$, $z_* = (2t_* - a - b)/\ell$. To see this, expand in the orthonormal basis $\sqrt{(2j+1)/\ell}\,L_j((2t-a-b)/\ell)$ and apply Cauchy–Schwarz. The bounds $|L_j(z)| \le 1$ on $[-1, 1]$ and $|L_j(z)| \le (|z| + \sqrt{z^2-1})^j$ outside follow from

$$L_j(z) = \frac{1}{\pi}\int_0^\pi (z + \sqrt{z^2-1}\cos v)^j\,\mathrm{d}v,$$

using the imaginary square root inside the interval and parity for negative z. The formula is a polynomial identity by expansion. At $t_* = -1$, $z_* = -3/(1-2\epsilon)$. First letting $n \to \infty$ and then $\epsilon \downarrow 0$ gives the matching lower exponent for both $\lambda_{n,s}$.

Finally use $I = [0, 1-\epsilon]$, where $\eta_2 \geq \epsilon/8$. The same evaluation estimate, uniformly over $t_* \in [0,1]$, gives

$$\|P_{n,0}\|_{[0,1]}^2 \leq \frac{8n^2 S_\epsilon^{2(n-1)}}{\epsilon(1-\epsilon)}\lambda_{n,2}, \quad S_\epsilon = s_\epsilon + \sqrt{s_\epsilon^2 - 1}, \quad s_\epsilon = \frac{1+\epsilon}{1-\epsilon}.$$

Integrating against η_4 bounds E_n by θ times this quantity. Now $S_\epsilon \to 1$ as $\epsilon \downarrow 0$; combined with $E_n \geq \lambda_{n,4}$ this proves the asserted rate. □

Write $r_n'(0) = p_n/A_n$ in lowest terms, $A_n > 0$. Its actual primitive form is

$$L_n = A_n\theta - p_n = A_n E_n, \qquad \log L_n = \log A_n - 2n\log\rho + o(n). \tag{9.4}$$

Along any proposed subsequence, $\limsup n^{-1}\log A_n < 2\log\rho$ is sufficient for shrinking values. The weak inequality is necessary, but equality requires control of the lower-order terms. No useful all-index reduced-denominator estimate has been established. The quadratic logarithmic decay of the raw determinant cannot be used as the scalar error rate: its target coefficient carries the same scale.

At $n = 2$, the first-variation pair is $50\theta - 49$ and the direct weight-four pair is $82\theta - 81$. The archived calculations cover exactly the direct and first-variation forms at $n = 1, \ldots, 12$, with full gcd removal. All twenty forms at $n = 3, \ldots, 12$ exceed one; the four at $n = 1, 2$ are smaller than one. Some diagnostic logarithms are shown below, rounded from the archived certified evaluations.

n	$\log A_n$	$\log E_n$	$\log L_n/n^2$
3	21.369976	−8.523832	1.427349
6	141.949801	−19.363907	3.405164
9	354.356711	−30.034542	4.003977
12	660.303908	−40.661541	4.303072

These are finite failures, not a growth theorem for A_n. An earlier table had cleared denominators without removing the common numerator factor; at $n = 12$ that overstated $\log L_n/n^2$ as 6.271803. The corrected data above are the retained conclusion. The archived arcsine realization, by contrast, was identified exactly with an earlier family; no such coefficient-sequence identification was established for this ordinary-moment variation.

10 The wider investigation, by method

This section reconciles the remaining records. Unless a result is proved above or in the companion paper, its status here is an *archived claim with the stated scope*, not a newly audited theorem. The register gives its definitions, original written arguments and recorded checks. It is deliberately more comprehensive than the selection of theorems.

10.1 Rational Euler and two-endpoint Hermite sources

These constructions cancel Taylor coefficients at one or two points. The aim is to force a small remainder while keeping its coefficients arithmetically manageable. Solving the interpolation

equations is only the first step; their endpoint values still need nonvanishing and primitive-size estimates.

Records H01–H03 extend (3.1) to two-endpoint sources. The rational source conditions reduce to a four-by-five linear system, with singular cases and the constant coefficient retained. The circular Hermite specialization has an archived eventual rank result and a signed error asymptotic proportional to $(-1)^n n^{5/2} \log^2 n\,((\sqrt{2}-1)/2)^n$. Its actual primitive height remains uncontrolled. Factorial rescaling of the entire pair does not change that question.

One arithmetic ingredient, H02, can be stated independently of that source. Products of r distinct integers in $[1, N]$, $N \geq r$, are all divisors of

$$W_r(N) = \prod_{j=1}^{r} \operatorname{lcm}(1, \ldots, \lfloor N/j \rfloor).$$

At a prime p the maximum possible product valuation is $\sum_{k\geq 1} \min(r, \lfloor N/p^k \rfloor)$; this is exactly $v_p(W_r(N))$. The maximum is attained by choosing the r integers with highest valuations, since the sets of multiples of p^k are nested. For the strict integration indices of the archived source this gives a sufficient coefficient denominator. In particular $\log W_4(N) \sim (1 + 1/2 + 1/3 + 1/4)N = 25N/12$ by the prime number theorem. It is not an endpoint primitive-value bound.

10.2 Continued fractions, derivatives and asymptotic summation

These methods improve a series tail by a rational approximation or an asymptotic correction. Differentiation can change the target constant, but also changes the denominator cost. It is essential to specify which endpoint and truncation parameters are allowed to grow together.

S04 and S05 are Theorems 4.1 and 5.1; their different quantifiers matter. Euler–Boole covers all endpoint and truncation schedules, whereas the continued-fraction theorem fixes the endpoint and restricts arithmetic to even depth. E05 concerns rational gauges of a particular ordinary two-state tail: the archived argument excludes its desired polynomial unimodular transfer. It says nothing about factorial gauges, higher-rank systems or unrelated recurrences. The binomial-shift reduction at the end of Section 5 also prevents counting an old scalar source as a new continued-fraction construction.

10.3 Modular and Eichler sources

Modular transformations relate expansions of a function at different points. The hope is to combine arithmetic information from those expansions with an analytic estimate improved by the symmetry. The experiments below test cancellation of unwanted constants, removal of selected prime denominators, and the singularities that remain.

M01 cancels an auxiliary Eichler period polynomial to produce a level-32 θ-only source. M02 constructs quadratic prime filters compatible with the Fricke transformation, obtaining selected savings in both endpoint germs. Neither transformation identities nor local savings alone establish favorable primitive arithmetic for growing filters. M03 records uncancelled interior poles in proposed higher-level coordinates and prefactors. A later annihilator removes ordinary and shifted pole families but leaves six fixed-point poles. These are limitations of those coordinates and raw coefficient estimates.

M04 compares three specified rational-system templates of ranks $9, 17, 18$. Its optimized displayed logarithmic derivative bound is below 1.3, against a required threshold above 4 in that test. This closes the prescribed comparison, not all modular coordinates. The exact template and source

choices are retained in the record; they are essential hypotheses, not an incidental implementation detail.

10.4 Polylogarithm ladders and functional equations

Functional identities replace one special value by combinations at other arguments. Small arguments improve series convergence, and shared arguments can permit joint cancellation. The remaining task is to control all coefficients and auxiliary logarithms in the resulting form.

L01 is the completed Broadhurst identity used in Section 7; its attribution is part of the statement. L02 studies equal-speed scalar Taylor truncations of one rational realization. For its partial-sum denominator q_N, the archived exact dyadic exponents are

$$v_2(q_{4m}) = 10m + 4, \quad v_2(q_{4m+1}) = 10m + 5, \quad v_2(q_{4m+2}) = 10m + 11, \quad v_2(q_{4m+3}) = 10m + 12,$$

with the first formula restricted to $m \geq 1$. Combined with the signed error estimate, these give a primitive obstruction for that scalar truncation, not for other uses of the ladder.

L03 uses the common base $(1 - i)/2$, moving arguments at powers $2, 3, 5$, rather than approximating the three endpoint values separately. The archived determinant has infinitely many nonzero evaluations and real decay compatible with its proved dyadic cost, leaving the margin $(\log 2)n^2/60$. Odd-prime arithmetic remains unresolved. Being small and 2-integral is not being a small integer form. The constrained positive source in Theorem 7.1 is a separate use of the same kernel and does not exclude this determinant.

L04 uses simultaneous pole multipliers sharing endpoint value 25^n. Its rational error has a nonzero asymptotic proportional to $(-1)^n 27^{-n}/\sqrt{n}$, but full primitive arithmetic is missing. L05 specializes Gangl's weight-four functional equation [6] to a 21-argument algebraic presentation. The archive checks the symbol identity and tracks completion terms. A naive trace after separate inversions fails; no theorem of numerical period independence follows from a symbol computation.

10.5 Slow dilations, prime deletion and sparse Padé

These attempts modify the generating series to reduce its arithmetic cost: spread out its exponents, remove contributions from selected primes, or retain a sparse set of terms. Such changes also alter the singularities and the available approximation estimates, so an arithmetic saving alone does not settle the construction.

A01 studies a fixed odd dilation parameter k and finds coefficient denominator rates between $3/2 + 1/(k + 2)$ and $3/2 + 1/k + 1/(2k + 4)$ in the normalization defined in its record. These are fixed-k coefficient statements, not primitive determinant rates or estimates uniform in a jointly growing k. A02 relates a bordered determinant's content to maximal-minor gcds and modular Padé defects. In its notation $T = DH$, $u = Db$, $d = \det T$, $m = \operatorname{adj}(T)u$, and $Q = u^T \operatorname{adj}(T)u$, the specialized gcds obey $\Delta \mid g \mid 6D\Delta^2$, where $g = \gcd(6Dd, Q)$ and $\Delta = \gcd(d, m_i)$. An example contains prime 23 absent from the coefficient denominator. The determinant identities are classical; the bounds do not settle all asymptotic primitive content.

A03 concerns failed fixed-cut absolute-contour estimates for $k = 3, 5$. It is not a lower bound on actual determinant size. A06 explains why a proposed same-variable positivity repair fails: the slow filter has genuine nonreal branch points satisfying $z^k = -(\sqrt{2} - 1)^2$, inconsistent with the proposed compact real-supported Cauchy transform after the analyzed algebraic repairs. Different changes of variable are outside that statement.

A04 deletes selected prime layers by Euler factors and can achieve subquadratic *sufficient* arithmetic cost for certain growing cutoffs. Required real decay and all-index nonvanishing are

absent. A05 identifies a keep-prime-3 sparse family with a simpler Padé sequence and proves a primitive obstruction only in an explicitly oversieved regime, whose cutoff prime is at least the fourth power of its size parameter. The natural cutoff was not excluded.

10.6 Positive determinants and interleaved frames

Changing the row and column nodes of a determinant may improve the balance between its real size and its prime denominators. The key distinction is between changing a basis within the same polynomial pencil, which preserves its primitive polynomial, and choosing a different set of rows, columns or moments, which can change the problem.

C01–C08 are the companion paper's positive cubic-tail kernel, uniform Cauchy comparison, consecutive-principal obstruction, difference-Gram construction and exact large-prime primitive arithmetic. Its two primitive lower exponents are $2-2\log 2$ and $6-\log 4$ on the n^2 scale. S08 is a separate archived obstruction for an overshifted dense grid with $m \geq 768r^4$; it does not address proportional shifts. The unrestricted source cone is covered in Section 2.

The nonprincipal experiment P06 was motivated by the interleaving and local pairing in Litsey's trigamma work [9]. For fixed $0 < c \leq 1$, let $m = \lfloor cr/2 \rfloor$, common nodes $C = \{1, \dots, r\}$, row nodes $C \cup \{r+1, r+3, \dots, r+2m-1\}$, and column nodes $C \cup \{r+2, r+4, \dots, r+2m\}$. Use the companion paper's kernel, replacing θ in the common diagonals by X. Its determinant $P_{r,c}(X)$ has degree r and positive evaluation at θ. The Cauchy comparison gives

$$\log P_{r,c}(\theta) = -E(c)r^2 + o(r^2), \quad E(c) = \iint \rho_c(x)\rho_c(y) \log \frac{x+y}{|x-y|}\,\mathrm{d}x\,\mathrm{d}y,$$

where $\rho_c = (\mathbf{1}_{[0,1]} + \mathbf{1}_{[0,1+c]})/2$. The archived four-layer local Smith calculation contributes $3(1+c^2)r^2 + o(r^2)$ to its sufficient arithmetic bound. It cannot beat the real exponent: positivity of the energy integrand and $\rho_c \leq 1$ imply

$$E(c) \leq 2(1+c)^2 \log 2 \leq 4(1+c^2)\log 2 < 3(1+c^2).$$

This is a limitation of that four-layer calculation throughout the width range, not an all-index primitive obstruction.

At $c = 1/5$ an actual coefficient valuation gives a narrower statement: every positive *integer* multiplier clearing coefficients produces growing values. Its lower clearing exponent is $2-c-c^2 = 1.76$, whereas $E(1/5) = 1.652307\ldots$. Rational primitive normalization can still divide common numerator content. The leading coefficient is a nonconstant rational border determinant, so the monic argument from the principal family cannot be transferred. The nine prescribed primitive examples fail after complete gcd removal, but no all-index bound on that content was proved. P07, developed in Section 9, is another open arithmetic problem rather than a consequence of P06.

10.7 Hypergeometric and quarter-pole constructions

Factorial ratios and hypergeometric identities can provide rational coefficients together with exact remainder formulas. Shifting or differentiating their parameters changes the special values that occur. The tests here seek a combination containing only θ and a rational term, then examine the denominator powers and common factors.

Q01 differentiates an integer-pole tail to isolate θ, with signed error proportional to

$$(-1)^{n+1} n^{-5/2} \log^2 n (27/256)^n.$$

Its displayed clearing multiplier retains fifth odd-prime layers; the resulting unfavorable bound is not a theorem about full primitive content. Q02 tests three nonsymmetric half-parameter jets and a 120-element Thomae orbit at one fixture. Its six primitive eliminations are finite evidence, not a classification of transformations.

Q03 supplies an all-index moved-base Whipple identity, a positive terminating coefficient expression, dyadic valuations and eventual signed nonvanishing. Its proved multipliers obey a divisibility relation, and integer clearing of the selected cofactor fails even before odd-prime costs. Unbounded odd common content remains outside the result. The symmetric candidate was exactly an earlier archived family.

The paired forward/dual mechanism in Leibtag et al. [8, Section 6.2] motivated Q04–Q05. An initial quarter-sampling change produced unequal numerators. The later construction deforms the pole grid, both parameter trajectories, initial states, readouts and normalization together, repairing that mismatch. For $\alpha = -3/4, -1/4$ its forward denominator is

$$Q_n = \sum_{k=0}^{14n} \binom{k+12n+\alpha}{10n}\binom{k+13n+\alpha}{10n}\binom{k+14n+\alpha}{10n}\binom{15n}{n+k}\binom{15n}{k}.$$

The full numerator formulas and dual coefficient rows are retained in the exact deformation record. The corrected all-index identities and ten telescoping certificates are archived mathematical and computational results; they were not rebuilt in this supplement pass.

The two arguments $a = 1/4$, $b = 3/4$ give the projected form

$$W_n = Q_n^a Q_n^b \theta - \frac{P_{4,n}^a Q_n^b - P_{4,n}^b Q_n^a}{256}.$$

The factor 256 comes from the difference of the quarter Hurwitz zeta values. The established relative rate and coefficient rate are $112.807539\ldots$ and $65.451536\ldots$. The specified termwise arithmetic bounds can certify at most $88.207529\ldots$ of the odd-prime content density, versus $142.862026\ldots$ required by that decay estimate. This ceiling does not bound all actual gcds, cancellation inside sums, or possible faster decay after projection. Projected nonvanishing was checked at only five indices. The repaired identity is retained; the earlier numerator mismatch is not a current reason to reject it.

10.8 Coupled integrals, geometry, and exact rediscoveries

Coupling several integrals or introducing extra parameters can cancel unwanted constants while retaining a positive expression. One must then check both the resulting arithmetic and whether the apparent new construction is actually a reparametrization of an earlier one.

P01 and O01 navigate the earlier lower-period cancellations, coupled holonomy, lattice, radial and orthogonal constructions. They are umbrella records, not single consolidated theorems. Some fitted recurrences were conjectural before later certification; those stages remain distinct.

P02 identifies the positive dimensional family exactly as $J_n = (n!)^4 4^{-n} L_{n+1,n+1}$ in the archived balanced-ladder notation. It is a rediscovery. P03 is a different nonuniform internal graph with $D = 8n + 4$, outer powers $n + 1$ and central Green power $4n + 1$. It has positive decay 256^{-n} and a terminating reduction to a five-period space for each fixed n. Its four tested pure-θ cofactors fail; no all-index primitive-height result follows.

P04 uses dense quarter poles $(2j + 1)/4$ and interpolation to cancel $\zeta(4)$, obtaining a positive Gram pencil with full target degree. Its analytic comparison with a single-pole kernel is not a primitive polynomial equivalence. P05 uses unsquared quarter poles. For $h = 2n$, $K = 8n + 4$ and

prime $p = 8n + 5$, its valuation $-8n$ under hypothetical p-integral rational θ proves conditional nonvanishing along admissible primes. Neither construction has a useful all-index primitive bound. Three and seven finite primitive cases, respectively, are not substitutes for it.

S07 is a quarter-Gamma fourth-cumulant extraction from the weighted projective data $(1, 3, 4, 4, 4, 4, 4)$. Its scalar sequence uses finite Hurwitz fourth-power sums, with weights $1, 3, 4, 4, 4, 4, 4$ and total 24, at the two quarter arguments. The fourth logarithmic coefficient is formed with all lower products retained. Its archived arithmetic proof gives reduced-denominator logarithm at least $24n + o(n)$, whereas the error is of order n^{-5}. That scalar obstruction does not certify the larger geometric interpretation or exclude the separate nonconstant deformation. No geometric theorem is needed for the finite-sum obstruction.

E09 records other exact collapses: signed divisibility/LCM Gram determinants become linear approximants by rank one, and complete Boolean bases reproduce an Euler product. Polynomial Dirichlet convolution retains the fourth denominator layer but also pays the hypothetical rational denominator to the polynomial degree. Untested bases are not excluded.

10.9 Euler functionals and proposed analytic bridges

Here the main issue is whether a proposed connection between two kinds of information is valid: for example, whether a p-adic functional estimate controls the relevant coefficients, or whether an analytic property supplies the arithmetic conclusion needed for irrationality. A counterexample to such a connection closes only the stated implication.

E01 computes a norm on integer-valued polynomials, rather than a Gauss coefficient norm. For the archived functional $\tau = -E_0 D^3/12$ and $p \geq 5$, the exact layer-onset function is $m_p(3q + r) = (3 - r)p^q + rp^{q+1}$, $0 \leq r < 3$; its first three positive layers start at $p + 2, 2p + 1, 3p$. This says nothing by itself about rational poles or primitive determinant content.

E02 applies Euler distribution to the whole polynomial-plus-pole functional, with the affine target change $Y = \chi_4(p)p^4 X + C_p$. The positive csch functional was already present in Carreira [4, Section 11]. Neither the distribution identity nor a formal target change equates real and p-adic limits. E03 supplies countermodels to particular proposed analytic implications, including an E-function Stokes inference, a compact-support inference from rational moments, and a spectral-raising bridge. Each countermodel has its stated hypotheses; none excludes the entire surrounding subject.

10.10 Cyclotomic parity and rationality-dependent constructions

The remaining tests use algebraic conjugation or begin by supposing θ rational. In the first case a trace can vanish by symmetry; in the second, an integrality claim or a differential construction must actually distinguish the hypothetical rational value. Neither property follows merely from writing down the corresponding formal expression.

The direct even-power signature extension E08 lies in a cyclotomic component negated by an involution fixing $\mathbb{Q}$, so its prescribed rational traces vanish. This parity statement does not apply to the nonlinear norm of Section 6; that family needed its own all-conjugate argument.

E07 assumes $\theta = u/v$ and tests the Niven-type source $[vF(u - vF)]^n/n!$, where $F(x) = \sum_{m \geq 1} \chi_4(m)x^m/m^4$. Its endpoint values vanish, but a derivative of order $p + n - 1$ at zero has valuation -3 for odd $p > n$ outside the fixed numerator factors. Thus the proposed universal endpoint-integrality step is false. Changing the integration variable to F gives a rational beta integral, not an irrationality proof. Selected finite jets or larger constructions are not covered.

E06 subtracts the prescribed polar terms of a rationality-dependent q-series. At $q = (p-1)/p$, its p^2 truncation has exact denominator p^{p^2} and growing numerator. This is a result for that rational

sampling and truncation, not for all q-series. E04 is the different monodromy theorem proved in Section 8. In both cases the hypothesis of rationality supplies no unspecified extra geometric or p-adic relation.

11 Computations, corrections, and review status

The repository separates three layers: the mathematical exposition, the inventory and method catalog, and verbatim research snapshots. The latter are historical evidence. A tentative conclusion inside an old note is not a current theorem if its later review corrected it. The register covers all 57 grouped entries and all twelve method groups; this is coverage of the selected archive, not a claim that every idea ever proposed for $\beta(4)$ has been investigated.

The companion paper's core exact and Arb checks have their own runner. For this supplement, a separate runner executes preserved scripts in a temporary directory, records fresh results, and checks the frozen source hashes. It includes the rational Sturm and tail certificate used in Theorem 7.1. Additional finite algebra checks cover the rewritten scalar, Euler–Boole, continuant and monodromy formulas. The precise cases, counts, dependencies and results are in `checks/supplement-reproduction.json`. No large parameter search is part of this writing pass. Archived tests outside that runner retain their original status and are not described as rerun.

Three corrections are particularly consequential. The quarter forward/dual identity mismatch was repaired in the later deformation record. The ordinary first-variation examples require removal of the *full* common numerator gcd; their original clearing-only table is superseded. Finally, the interleaved-frame negative Smith layers and integer-clearing obstruction do not bound every rational primitive rescaling. The final method record preserves these distinctions.

All work was prepared with Codex. The historical record contains separate model cross-checks as well as calculations. This supplement pass is a review by its assembling model, not a new independent model audit, human mathematical audit or proof-assistant verification. Finite exact calculations check identities and specified instances; they do not prove asymptotic statements by extrapolation. The Sturm and tail calculation is explicitly distinguished because it certifies finite inequalities used in one proof.

The source ledger records versions, local PDFs, hashes, actual reading scope and unchecked dependencies. The specific obstruction statements are candidates for external scrutiny, not priority claims inferred from a failed literature search. Classical positivity, interpolation, orthogonal-polynomial, continued-fraction and content-removal mechanisms remain classical even when applied in a new experiment.

A Claim and source register

Each entry gives the archived classification, the scope of this writing pass, and one primary navigation source. All paths are relative to the repository root. The complete proof and calculation path lists, limitations and literature links are in `planning/theorem-inventory.json`. The frozen archive classification does not mean every claim has been independently re-audited.

C01 – A positive cubic-tail kernel isolating beta(4) positive construction. Companion-paper proof; not repeated here.

`evidence/research/dilations-and-positive-kernels.txt`

C02 – Uniform two-sided comparison with a Cauchy determinant analytic theorem. Companion-paper proof; not repeated here.

`evidence/research/collaboration/round-01/construction/findings.txt`

C03 – Primitive growth for every consecutive principal offset primitive value obstruction. Companion-paper proof; not repeated here.

`evidence/research/collaboration/round-01/construction/findings.txt`

C04 – Difference-variable Gram moments with full target rank positive construction. Companion-paper proof; not repeated here.

`evidence/research/collaboration/round-06/fresh/findings.txt`

C05 – Matching real upper and lower rates for the difference Gram analytic theorem. Companion-paper proof; not repeated here.

`evidence/research/collaboration/round-06/fresh/findings.txt`

C06 – Exact large-prime primitive exponent for the difference Gram arithmetic theorem. Companion-paper proof; not repeated here.

`evidence/research/collaboration/round-06/fresh/findings.txt`

C07 – Primitive growth for the difference-variable Gram family primitive value obstruction. Companion-paper proof; not repeated here.

`evidence/research/collaboration/round-06/fresh/findings.txt`

C08 – Primitive invariance under full rational basis changes standard fact. Companion-paper proof; not repeated here.

`evidence/research/collaboration/round-01/construction/findings.txt`

S01 – The complete cone of scalar positive polynomial forms structural theorem. Unrestricted cone and degree law proved here; fixed-degree boundary formulas remain in the archive.

`evidence/research/collaboration/round-01/fresh/findings.txt`

S02 – Finite positive-cone extremizers do not exclude interior forms finite evidence. Catalogued with its archived scope; not promoted to a newly audited theorem.

`evidence/research/collaboration/round-01/fresh/findings.txt`

S03 – Repeated quadratic positive Mellin sources primitive value obstruction. Repeated-quadratic obstruction proved here.

`evidence/research/collaboration/round-03/fresh/findings.txt`

S04 – Uniform failure of the Euler-Boole truncation family primitive value obstruction. Uniform Euler–Boole obstruction proved here.

`evidence/research/collaboration/round-05/fresh/findings.txt`

S05 – Third-derivative continued fractions at fixed endpoints primitive value obstruction. Fixed-endpoint even-depth obstruction proved here.

`evidence/research/collaboration/round-07/fresh/findings.txt`

S06 – Cyclotomic minimal-polynomial obstruction primitive value obstruction. Prime-level sine-normalized obstruction proved here.

`evidence/research/collaboration/round-11/norms/findings.txt`

S07 – Quarter-Gamma cumulant sequence primitive value obstruction. Catalogued with its archived scope; not promoted to a newly audited theorem.

`evidence/research/collaboration/round-09/orbifold/findings.txt`

S08 – Overshifted dense grids primitive value obstruction. Catalogued with its archived scope; not promoted to a newly audited theorem.

`evidence/research/collaboration/round-04/coordinator/loewner_shift.txt`

S09 – A positive-source obstruction for the common-base Mellin kernel primitive value obstruction. Rational-kernel inequality proved here with exact certificates; beta identification uses the declared Broadhurst input.

`evidence/research/collaboration/round-03/fresh/mellin_positive_obstruction.txt`

A01 – Delayed prime layers in slowly varying dilations arithmetic theorem. Catalogued with its archived scope; not promoted to a newly audited theorem.

`evidence/research/collaboration/round-01/arithmetic/findings.txt`

A02 – Bordered determinant content and modular defects arithmetic theorem. Catalogued with its archived scope; not promoted to a newly audited theorem.

`evidence/research/collaboration/round-02/arithmetic/findings.txt`

A03 – Fixed-cut capacity limitations estimate limitation. Catalogued with its archived scope; not promoted to a newly audited theorem.

`evidence/research/collaboration/round-02/analysis/findings.txt`

A04 – Prime deletion with subquadratic sufficient arithmetic cost partial construction. Catalogued with its archived scope; not promoted to a newly audited theorem.

`evidence/research/collaboration/round-04/arithmetic/findings.txt`

A05 – Sparse collapse and an oversieved primitive obstruction primitive value obstruction. Catalogued with its archived scope; not promoted to a newly audited theorem.

`evidence/research/collaboration/round-04/construction/findings.txt`

A06 – Nonreal branch points obstruct a particular positivity repair structural obstruction. Catalogued with its archived scope; not promoted to a newly audited theorem.

`evidence/research/collaboration/round-01/construction/findings.txt`

H01 – Rational Euler/Hermite source reduction positive construction. Euler identity and one-endpoint reduction proved here; two-endpoint system retained as an archived claim.

`evidence/research/collaboration/round-03/fresh/findings.txt`

H02 – Strict integration denominator word arithmetic theorem. Strict-index denominator word proved here; source-specific sharpness retained as an archived claim.

`evidence/research/collaboration/round-01/arithmetic/findings.txt`

H03 – Circular Hermite sources: eventual nonvanishing and real error partial construction. Catalogued with its archived scope; not promoted to a newly audited theorem.

`evidence/research/collaboration/round-07/analysis/rank_asymptotic.txt`

L01 – Completed Broadhurst weight-four ladder attributed identity. Catalogued with its archived scope; not promoted to a newly audited theorem.

`evidence/research/collaboration/round-02/fresh/findings.txt`

L02 – Scalar Taylor truncations of the ladder primitive value obstruction. Catalogued with its archived scope; not promoted to a newly audited theorem.

`evidence/research/collaboration/round-02/fresh/findings.txt`

L03 – Common-base ladder determinant partial construction. Catalogued with its archived scope; not promoted to a newly audited theorem.

`evidence/research/collaboration/round-03/analysis/findings.txt`

L04 – Matched simultaneous multipliers partial construction. Catalogued with its archived scope; not promoted to a newly audited theorem.

`evidence/research/collaboration/round-02/fresh/simultaneous.txt`

L05 – Gangl specialization and completion/trace limitations attributed identity. Catalogued with its archived scope; not promoted to a newly audited theorem.

`evidence/research/collaboration/round-06/identities/findings.txt`

M01 – Eichler period cancellation and the level-32 source positive construction. Catalogued with its archived scope; not promoted to a newly audited theorem.

`evidence/research/modular-detour.txt`

M02 – Fricke-compatible prime filters arithmetic theorem. Catalogued with its archived scope; not promoted to a newly audited theorem.

`evidence/research/collaboration/round-05/arithmetic/findings.txt`

M03 – Interior poles in higher-level coordinate repairs structural obstruction. Catalogued with its archived scope; not promoted to a newly audited theorem.

`evidence/research/collaboration/round-06/fresh/pole_annihilator_review.txt`

M04 – A bounded level-32 rational-system comparison estimate limitation. Catalogued with its archived scope; not promoted to a newly audited theorem.

`evidence/research/collaboration/round-13/findings.txt`

Q01 – Differentiated integer-pole tail partial construction. Catalogued with its archived scope; not promoted to a newly audited theorem.

`evidence/research/collaboration/round-04/fresh/findings.txt`

Q02 – Nonsymmetric Whipple jets and one Thomae orbit finite evidence. Catalogued with its archived scope; not promoted to a newly audited theorem.

`evidence/research/collaboration/round-10/symmetry/findings.txt`

Q03 – Moved-base Whipple pairing estimate limitation. Catalogued with its archived scope; not promoted to a newly audited theorem.

`evidence/research/collaboration/round-14/findings.txt`

Q04 – Repaired paired forward/dual quarter identity positive construction. Catalogued with its archived scope; not promoted to a newly audited theorem.

`evidence/research/collaboration/round-12/fresh/findings.txt`

Q05 – Arithmetic ceiling for the proved paired-identity estimates estimate limitation. Catalogued with its archived scope; not promoted to a newly audited theorem.

`evidence/research/collaboration/round-15/coordinator/arithmetic-gate.txt`

P01 – Lower-period cancellation in earlier positive/coupled integrals partial construction. Catalogued with its archived scope; not promoted to a newly audited theorem.

`evidence/research/structural-followup.txt`

P02 – Dimensional ladder equals an archived balanced ladder equivalence. Catalogued with its archived scope; not promoted to a newly audited theorem.

`evidence/research/collaboration/round-08/coordinator/dimensional-ladder-gate.txt`

P03 – Nonuniform internal graph: fixed-period space and positive decay partial construction. Catalogued with its archived scope; not promoted to a newly audited theorem.

`evidence/research/collaboration/round-09/internal_graph/general-period-space.txt`

P04 – Positive dense-quarter interpolation partial construction. Catalogued with its archived scope; not promoted to a newly audited theorem.

`evidence/research/collaboration/round-10/coordinator/dense-quarter-gate.txt`

P05 – Unsquared quarter poles: conditional nonvanishing partial construction. Catalogued with its archived scope; not promoted to a newly audited theorem.

`evidence/research/collaboration/round-11/fresh/findings.txt`

P06 – Interleaved nonprincipal frames: exact local layers, insufficient bound estimate limitation. Catalogued with its archived scope; not promoted to a newly audited theorem.

`evidence/research/collaboration/round-16/archive-audit.txt`

P07 – Ordinary-moment first variation: positivity but open arithmetic partial construction. Analytic sign and rate proved here; primitive arithmetic remains open.

`evidence/research/collaboration/round-18/findings.txt`

E01 – Sharp Euler functional norm on integer-valued polynomials arithmetic theorem. Catalogued with its archived scope; not promoted to a newly audited theorem.

`evidence/research/collaboration/round-10/adelic/findings.txt`

E02 – Whole-functional Euler distribution attributed identity. Catalogued with its archived scope; not promoted to a newly audited theorem.

`evidence/research/collaboration/round-08/discrete/findings.txt`

E03 – Scoped countermodels for proposed analytic bridges structural obstruction. Catalogued with its archived scope; not promoted to a newly audited theorem.

`evidence/research/coordinated-round-08.txt`

E04 – Monodromy obstruction to a linear endpoint regularization structural obstruction. Scoped monodromy statement proved here.

`evidence/research/collaboration/round-12/geometry/findings.txt`

E05 – Rational gauges of the ordinary two-state tail structural obstruction. Catalogued with its archived scope; not promoted to a newly audited theorem.

`evidence/research/collaboration/round-12/fresh/tail_gauge_gate.txt`

E06 – A rationality-dependent q-series truncation primitive value obstruction. Catalogued with its archived scope; not promoted to a newly audited theorem.

`evidence/research/collaboration/round-12/arithmetic/findings.txt`

E07 – Endpoint integrality fails for a particular Niven source structural obstruction. Catalogued with its archived scope; not promoted to a newly audited theorem.

`evidence/research/collaboration/round-11/rationality/findings.txt`

E08 – Direct even-power signature transfer has zero rational trace structural obstruction. Catalogued with its archived scope; not promoted to a newly audited theorem.

`evidence/research/collaboration/round-10/fresh/findings.txt`

E09 – Multiplicative and divisibility-Gram rediscoveries equivalence. Catalogued with its archived scope; not promoted to a newly audited theorem.

`evidence/research/collaboration/round-09/independent/multiplicative_gate.txt`

O01 – Earlier coupled holonomy and lattice explorations partial construction. Catalogued with its archived scope; not promoted to a newly audited theorem.

`evidence/research/coupled-holonomy.txt`

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